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LAB ASSIGNMENT-8

Border Gateway Protocol (BGP)

Title: Configuration and Analysis of BGP Routing

1. Objective

- To understand the working of **Border Gateway Protocol (BGP)**
- To configure **External BGP (eBGP)** between different Autonomous Systems
- To analyze **path selection using BGP attributes**

2. Theory

Border Gateway Protocol (BGP) is a **path vector routing protocol** used to exchange routing information between different **Autonomous Systems (AS)** on the Internet.

Key Concepts

- **Autonomous System (AS):** A group of networks under a single administrative domain
- **eBGP:** Routing between different AS
- **iBGP:** Routing within the same AS
- **Path Attributes:** Used to select best route
  - AS_PATH
  - NEXT_HOP
  - LOCAL_PREF
  - MED

BGP Characteristics

- Uses **TCP port 179**
- Supports **policy-based routing**
- Ensures **loop-free routing** using AS_PATH

3. Topology

Description Network

Setup:

- 3 Routers (R1, R2, R3)
- Each router belongs to a different AS

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- R1 → AS 100
- R2 → AS 200
- R3 → AS 300

- Each router connected to a LAN

4. IP Addressing Scheme

Device	Interface	IP Address	Subnet Mask
R1	G0/0	192.168.1.1	255.255.255.0
R1	G0/1	10.0.0.1	255.255.255.252
R2	G0/0	10.0.0.2	255.255.255.252
R2	G0/1	10.0.0.5	255.255.255.252
R3	G0/0	10.0.0.6	255.255.255.252
R3	G0/1	192.168.3.1	255.255.255.0

5. Procedure

Step 1: Create Topology

- Open Cisco Packet Tracer
- Place 3 routers and connect using serial/ethernet links
- Add PCs to each LAN

Step 2: Configure IP Addresses

Configure IP addresses on all router interfaces.

Example (R1):

```
conf t
interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown

interface g0/1
```

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```
ip address 10.0.0.1 255.255.255.252
no shutdown
```

Step 3: Configure BGP

R1 Configuration (AS 100)

```
router bgp 100
neighbor 10.0.0.2 remote-as 200
network 192.168.1.0 mask 255.255.255.0
```

R2 Configuration (AS 200)

```
router bgp 200
neighbor 10.0.0.1 remote-as 100
neighbor 10.0.0.6 remote-as 300
network 10.0.0.0 mask 255.255.255.252
```

R3 Configuration (AS 300)

```
router bgp 300
neighbor 10.0.0.5 remote-as 200
network 192.168.3.0 mask 255.255.255.0
```

Step 4: Verify Configuration

Use the following commands:

```
show ip bgp
show ip route
show ip bgp summary
```

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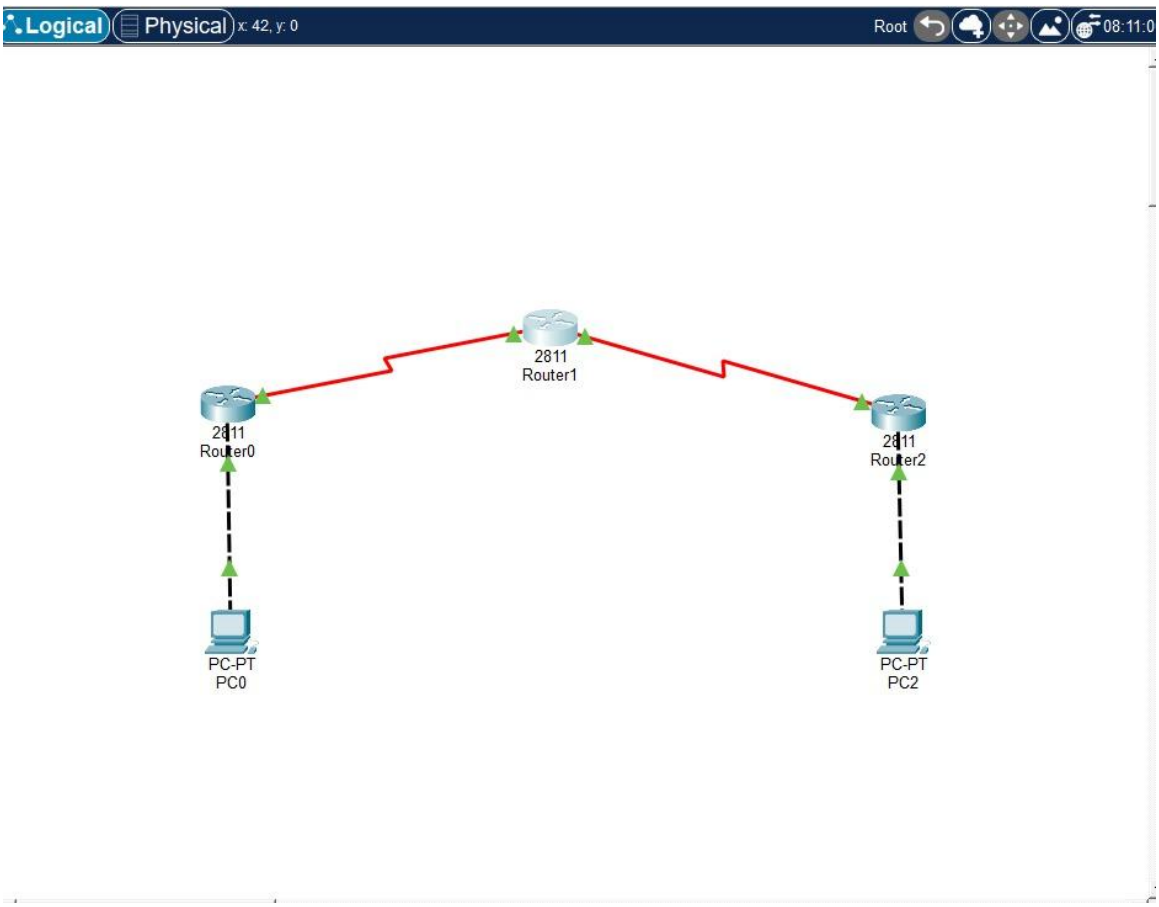
Solution